

An Adaptive Overcurrent Protection Scheme for Dual-Setting Directional Recloser and Fuse Coordination in Unbalanced Distribution Networks With Distributed Generation

Muhammad Yousaf¹, Student Member, IEEE, Amin Jalilian², Student Member, IEEE, Kashem M. Muttaqi³, Senior Member, IEEE, and Danny Sutanto⁴, Senior Member, IEEE

Abstract—The recloser–fuse coordination of the distribution networks (DNs) equipped with distributed generators may be jeopardized by using the conventional overcurrent protection strategies thus degrading the network reliability. This article focuses on the design of a protection coordination strategy based on the dual-setting directional recloser (DSDR) for the effective implementation of a fuse-saving scheme in DNs. The proposed DSDR-based protection strategy will assist the network operators by providing a flexible coordination range along with more accurate protection settings to meet the protection coordination requirements. The performance of the proposed protection scheme has been validated through its implementation on IEEE standardized test systems and a comparative analysis has been drawn to demonstrate its effectiveness. Furthermore, this article concentrates on the economical design of the protection strategy for DNs by utilizing fewer numbers of protective devices. The results show that the proposed strategy will considerably reduce the number of temporary and sustained interruptions experienced by the consumers of large DNs thus improving the network reliability indices.

Index Terms—Distributed generation, distribution networks (DNs), fault studies, overcurrent protection, protective devices (PDs), recloser–fuse coordination.

I. INTRODUCTION

THE proliferation of distributed energy resources (DERs) in the power networks offers many advantages for the customers and utilities such as power quality and reliability enhancement [1], [2]. Despite the advantages of DERs, the

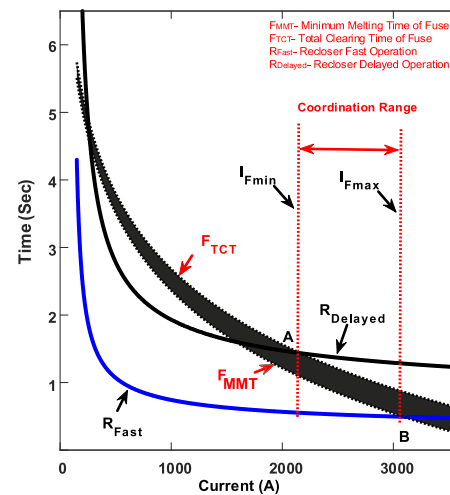


Fig. 1. Conventional time-current characteristics (TCC) of the recloser–fuse coordination to implement the fuse-saving scheme.

deployment of the DERs can originate several control and protection challenges to modern distribution grids [3]. Subsequent to large-scale integration of distributed generator (DG) units, the conventional concept of unidirectional power flow is revisited [4], followed by the evolution in the operational practices for efficient power network operation under all scenarios [5].

A critical protection challenge caused by the integration of DERs is the loss of the recloser–fuse coordination, which may seriously degrade the selectivity and reliability indices of the protection network [6], [7]. In distribution grids, the recloser–fuse coordination is essentially intended to implement the fuse-saving scheme to prevent the conversion of the momentary interruptions into sustained ones [8]. The recloser–fuse coordination is valid only if the fault current lies in between the permissible coordination range between minimum (I_{Fmin}) and maximum (I_{Fmax}) fault current as demonstrated by points A and B in Fig. 1 [9]. The recloser fast operation (R_{Fast}) is set to operate before the minimum melting time (MMT) of the associated fuse (F_{MMT}) to prevent the fuse melting for the self-clearing of momentary faults [10]. For persistent faults, the recloser lockout ($R_{Delayed}$) operation will clear the faulted section if the fault is

Manuscript received July 31, 2021; revised October 7, 2021; accepted October 14, 2021. Date of publication January 25, 2022; date of current version March 20, 2022. Paper 2021-PSPC-0758.R1, presented at the 2021 IEEE Industry Applications Society Annual Meeting, Vancouver, BC Canada, Oct. 10–14, 2021, and approved for publication in the IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS by the Power Systems Protection Committee of the IEEE Industry Applications Society. This work was supported in part by the Higher Education Commission, Pakistan, and in part by the University of Wollongong, Wollongong, NSW, Australia under the UOW–HEC Joint Scholarship Program. (Corresponding author: Muhammad Yousaf.)

The authors are with the Australian Power Quality and Reliability Centre, School of Electrical, Computer and Telecommunications Engineering, University of Wollongong, Wollongong, NSW 2522, Australia (e-mail: my701@uowmail.edu.au; aj085@uowmail.edu.au; kashem@uow.edu.au; dsutanto@uow.edu.au).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TIA.2022.3146095>.

Digital Object Identifier 10.1109/TIA.2022.3146095

not removed by the total clearing time (TCT) of the fuse (F_{TCT}), i.e., the fuse operation [11].

However, authors in [12] have concluded that the recloser fast operation can be marginally revised to counter the increased fault current scenarios in the presence of DERs to retain the recloser–fuse coordination. In [13], the instantaneous overcurrent (51 and 50) elements were utilized to an extent for the recloser fast operation to attain the desirable coordination time interval (CTI) among the protective devices (PDs). Conversely, the instantaneous operation during the recloser fast mode may also instigate coordination issues for the other PDs specifically those located upstream of the mid-line recloser due to the reverse flow of fault current [14].

In [15], the authors also utilized superconducting fault current limiter to marginally suppress the fault level variance to retain the recloser fuse coordination in the distribution network (DN), which may also introduce additional cost and maintenance limitations. The islanded operation of DGs can also improve the reliability indices of the DNs as suggested in [16], while the recloser coordination issue will be subjected to undesirable DG capacity and location constraints.

Furthermore, the adaptive relaying strategy has also been proposed for the steadfast detection of fault by the continuous modification of the relay pickup setting in real-time operations based on the available system information [17], [18]. Because of the recent advancement in the relay design, the concept of dual-setting relays has been recently reported to minimize the relay response time during faults, which can provide a substantial flexible range for the relay settings for coordination purposes [19], [20]. Moreover, the characteristics modification of dual-setting directional relays was also selected to optimize the relay coefficients along with additional relaying settings to restore the lost protection coordination in meshed distribution grids equipped with multiple DGs units [21]. To date, the concept of the dual-setting directional recloser (DSDR) has not been extended for its applicability to the recloser–fuse coordination in DNs.

In conjunction with the authors' earlier article presented in [22], the major contributions of this extended article are as follows:

- 1) Development of a novel protection strategy utilizing the DSDR to restore the recloser–fuse coordination.
- 2) To demonstrate the effectiveness of the DSDR deployment for forward and reverse directions.
- 3) Enhancement of the flexible coordination range of recloser–fuse coordination to improve network reliability.
- 4) To minimize the requisite PDs and power disruption utilizing the DSDR.
- 5) To validate the performance of the proposed protection strategy on standardized test feeders.

In addition, the dual-setting directional reclosing unit will provide two pairs of adjustable settings both for forward and reverse direction, i.e., dual time-dial settings (TDS) and dual pick-up current settings. Due to the diverse short-circuit capacity of the grid and DG units, the fault current experienced by the recloser will be dissimilar in the forward and reverse direction thus requiring different pickup and TDS settings to establish

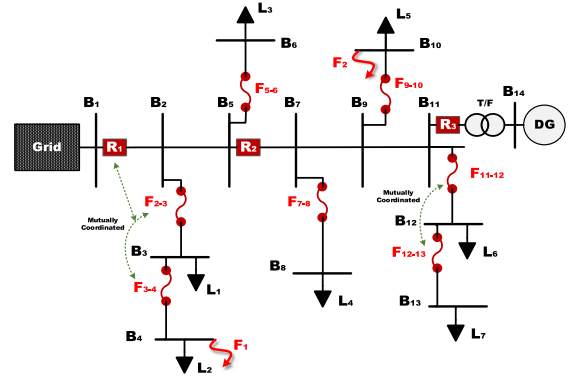


Fig. 2. Configuration of a typical distribution network equipped with reclosers and fuses for safe operation.

effective coordination. Therefore, this article focuses on the development of a protection coordination strategy for a DSDR and fuses for the improved flexible coordination range, specifically for the coordination of the series fuses on lengthy laterals.

II. PROBLEM FORMULATION

The essential PDs in a typical DN are the reclosers and fuses, which are mutually coordinated for the selective operation to clear the faults as demonstrated in Fig. 2.

In a typical DN as shown in Fig. 2, the recloser R_1 and both series fuses F_{2-3} and F_{3-4} will experience the same fault current for fault F_1 at bus B_4 in the absence of DG. The fast operation of the R_1 will de-energize the faulted section to self-extinguish the momentary fault. For a sustained interruption, the fault will be cleared by fuse F_{3-4} , while the operation of fuse F_{2-3} and the delayed operation of the R_1 constitute the secondary and tertiary layers of the protection network. Similarly, both reclosers R_1 and R_2 are mutually coordinated for a selective operation for the fault at F_2 to minimize the interruption to the customers.

After the DG incorporation at B_{11} , the power flow is not unidirectional and the traditional protection system cannot be very effective for all the possible fault locations. For the same fault F_1 at bus B_4 , both reclosers R_1 and R_2 should be coordinated with the associated series fuses F_{2-3} and F_{3-4} to completely de-energize the faulted section for the self-clearance of the temporary faults. However, the recloser R_2 is operating in the reverse direction, which experiences different fault levels. Thus, the recloser R_2 requires a new pickup and TDS setting based on the short-circuit capacity of the contributing source. The fast operation of both the reclosers R_1 and R_2 are mutually coordinated for the downstream faults such as fault at point F_2 to prevent the temporary interruption to the loads L_1 and L_2 to improve the momentary average interruption frequency index (MAIFI) of the system. However, bidirectional fault current experienced by the recloser R_2 demonstrates the intensity of this protection issue. Therefore, the DSDR will be required to employ to address this protection challenge. Furthermore, the installation of a DSDR will reduce the number of PDs installed, thus reducing the protection system complexities.

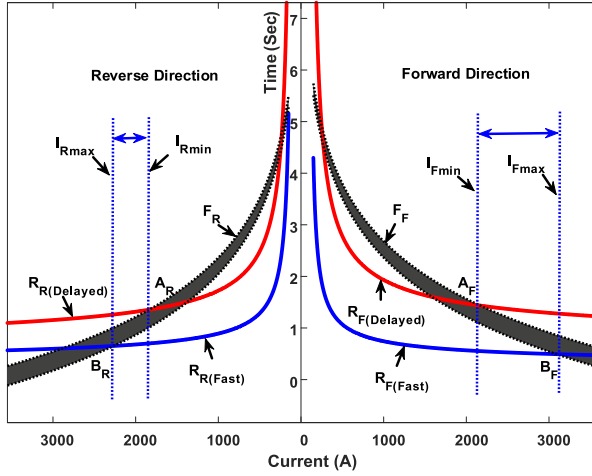


Fig. 3. Proposed time-current characteristics (TCC) for dual-setting recloser for the fuse-saving scheme.

A. Mathematical Formulations

The DSDR will be operating independently in both the forward and reverse direction as shown in Fig. 3 for the recloser-fuse coordination scenario. To formulate the fundamental mathematical model of this protection strategy, the operating time (t) for the fast operation (F) of the DSDR can be expressed in the forward direction (fw) as follows [23], [24]:

$$t_{k,l,F}^{fw} = TDS_{k,F}^{fw} \left(\frac{A_k^{fw}}{\left(\frac{I_{f,l}}{I_{p,k}} \right)^{n_k^{fw}} - 1} + B_k^{fw} \right) \quad (1)$$

where A_k^{fw} , B_k^{fw} , and n_k^{fw} are the relay constant for specific operating curve characteristics and " n_k " is the DSDR identifier for a fault location " l ." Moreover, I_f and I_p denote the relay fault current and pickup setting of recloser, respectively. Likewise, the operating time (t) for the delayed operation (D) of the recloser can be derived in the forward direction (fw) as follows [25], [26]:

$$t_{k,l,D}^{fw} = TDS_{k,D}^{fw} \left(\frac{A_k^{fw}}{\left(\frac{I_{f,l}}{I_{p,k}} \right)^{n_k^{fw}} - 1} + B_k^{fw} \right) \quad (2)$$

Furthermore, the forward pickup current setting of the DSDR for an overload factor (OLF) can be expressed as follows [12], [27]:

$$I_{p,k}^{fw} = OLF \times I_{nom,k}^{fw} \quad (3)$$

where $I_{nom,k}^{fw}$ represents the nominal load current during the steady-state operation for the k th recloser. Moreover, the operating constraints of k th reclosing PD can be subjected to the following constraints of TDS and fault ratings in the forward direction (fw) [19]:

$$TDS_{k,min} < TDS_{k,F}^{fw}, TDS_{k,D}^{fw} < TDS_{k,max} \quad (4)$$

$$I_{p,k,min} < I_{p,k}^{fw} < I_{p,k,max} \quad (5)$$

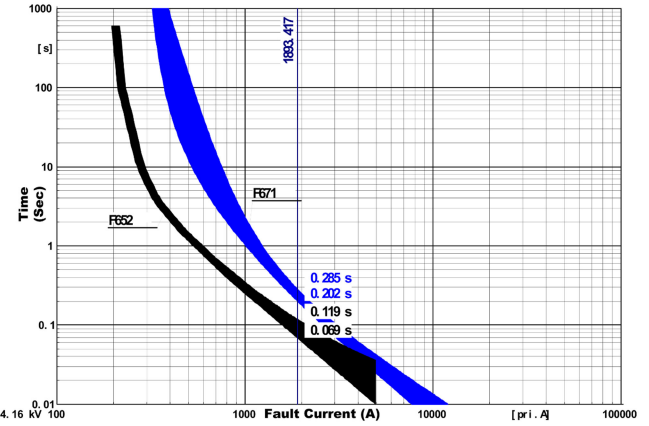


Fig. 4. Series fuse coordination for single line to ground (SLG) fault.

Similarly, the fuse operating time can also be determined using the fuse TCC curves comprising both MMT and the TCT curves expressed by the slope of a straight line of $I^2 t \log - \log$ plot as in (6) [25]:

$$\log(t_{fuse(i)}) = a_i \times \log(I_{f,i}) + b_i \quad (6)$$

whereas $t_{fuse(i)}$ is the operating time of the i th fuse experiencing a fault current $I_{f,i}$. The coordination fuse coefficients of the i th fuse are represented by a_i and b_i in (6). The value of the coefficient a_i should be selected the same for all the fuses in the DN because all the fuses will be of a similar type in a DN for mutual coordination purposes. Moreover, it also presents the slope of the straight line of the $I^2 t \log - \log$ plot curve of the fusing PD, whereas the coefficient b_i is computed through the coordination study for the fuse-saving scheme. Thus, the subsequent fuse coefficient b_i can be computed by determining its operating time using the average recloser fast and slow response time for a particular fault current (I_f) by using (7) [12]:

$$b_i = \log \left(\frac{t_{k,F}^{fw} + t_{k,D}^{fw}}{2} \right) - a_i \times \log(I_{f,i}) \quad (7)$$

The fuse-saving scheme should also be selective to the series fuses to minimize the power disruption against sustained interruptions. For the series fuses coordination, usually, 75% rule is generally adopted which states that 75% of the MMT of the upstream fuse should be higher than the TCT of the downstream fusing PD [28], [29]. This series fuses coordination rule can also be expressed as follows [28], [29]:

$$t_{fuse(i)}^{MMT} > 1.33 \times t_{fuse(i+1)}^{TCT} \quad (8)$$

The distribution fuses usually require three to five cycles to completely clear the faults predominantly depending on the fuse specifications as shown in Fig. 4. If the lengthy laterals have multiple series fuses such as $B_2 - B_4$ and $B_{11} - B_{13}$ as demonstrated in Fig. 2, (7) can be further modified to find the fuse coefficient b_i for the i th fuse in the faulted path to preserve the protection selectivity among different PDs as given in (9)

[25]:

$$b_i = \log \left(t_{k,F}^{\text{fw}} + \frac{i \times (t_{k,D}^{\text{fw}} - t_{k,F}^{\text{fw}})}{z + 1} \right) - a_i \times \log(I_{f,i}) \quad (9)$$

where z is the total number of series fuses in the faulted path and i represents the fuse number closest to the fault location.

For the reverse operating conditions as depicted by the fault at point F_1 in Fig. 2, R_2 will be subjected to operate in the reverse direction in the presence of DG. Thus, the operating time (t) of the k th reclosing PD can be expressed in the reverse direction (rv) for both fast (F) and delayed (D) operation as [20], [30]:

$$t_{k,l,F}^{\text{rv}} = \text{TDS}_{k,F}^{\text{rv}} \left(\frac{A_k^{\text{rv}}}{\left(\frac{I_{f,m}}{I_{p,k}^{\text{rv}}} \right)^{n_k^{\text{rv}}} - 1} + B_k^{\text{rv}} \right) \quad (10)$$

$$t_{k,l,D}^{\text{rv}} = \text{TDS}_{k,D}^{\text{rv}} \left(\frac{A_k^{\text{rv}}}{\left(\frac{I_{f,m}}{I_{p,k}^{\text{rv}}} \right)^{n_k^{\text{rv}}} - 1} + B_k^{\text{rv}} \right) \quad (11)$$

where A_k^{rv} , B_k^{rv} , and n_k^{rv} represent the specified characteristics curves of the DSDR in the reverse direction for an upstream fault location “ m .” Correspondingly, the pickup current and TDS setting constraints can also be revamped accordingly as follows [12], [19]:

$$I_{p,k}^{\text{rv}} = \text{OLF} \times I_{\text{nom},k}^{\text{rv}} \quad (12)$$

$$\text{TDS}_{k,\min} < \text{TDS}_{k,F}^{\text{rv}}, \text{TDS}_{k,D}^{\text{rv}} < \text{TDS}_{k,\max} \quad (13)$$

$$I_{p,k,\min} < I_{p,k}^{\text{rv}} < I_{p,k,\max} \quad (14)$$

where $I_{p,k}^{\text{rv}}$ denotes the pickup current setting of the k th reclosing PD in the reverse direction for an upstream fault. While $\text{TDS}_{k,F}^{\text{rv}}$ and $\text{TDS}_{k,D}^{\text{rv}}$ represent the recloser TDS for the fast and delayed operation, respectively. As the distribution fuses are nondirectional PDs, thus it does not require the upgrades of the fuses parameters. Therefore, the recloser–fuse coordination can be established for all the fault locations by utilizing the flexible coordination range as demonstrated in Fig. 3 for both directions. Similarly, the proper fuse–fuse coordination can be ensured for the improved selectivity of the protection network.

III. PROPOSED METHODOLOGY

The essential features of a protection scheme are to respond promptly and securely to the electrical faults in a suitable time to prevent damage to the power network components. This section describes an overview of the proposed scheme for the implementation of recloser–fuse coordination by utilizing the DSDR in DNs. The flow chart of the proposed protection strategy is shown in Fig. 5 for the sequential design of the fuse-saving scheme to improve the reliability of the protection network. The outlines of the proposed protection scheme are summarized as follows:

- 1) The load flow analysis will provide the assessment information of the rated current at different sites of the DN in the presence of the DGs at the optimum location and penetration level.

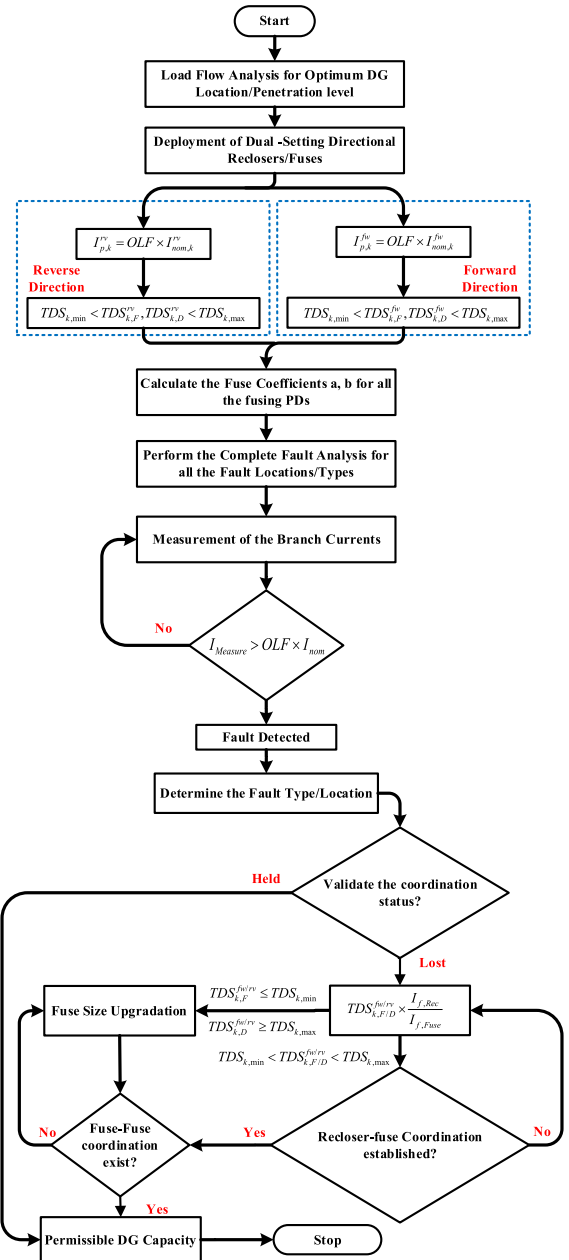


Fig. 5. Proposed methodology for the recloser–fuse coordination using DSDR in distribution networks.

- 2) The deployment of the DSDR and the fuses will be subjected to the formation of protection zones for each network section to minimize the consumer interruptions against any fault. In addition, the total number of PDs is kept at a minimum without compromising the network reliability as each additional PD will add complications and cost to the designed protection network.
- 3) The optimal TDS and pickup current settings are defined for all the reclosers for the forward and reverse operation. The fast operation of the DSDRs is mutually coordinated in the same direction to minimize the MAIFI index for the reliability improvement of the DN.

- 4) The fuse coefficients a_i and b_i are computed for all the fuses using (6)–(9) as explained in Section II to implement the fuse-saving scheme. The coordination among the series fuses on long laterals is also ensured for the improved selectivity for each protection zone.
- 5) The complete fault analysis is performed to assess the fault level data for different types of faults in the presence of DGs. The fault current contribution can also be observed from each source. The pickup current settings for the PDs of each protection zone should be less than the minimum fault current for that particular zone for the steadfast operation of the overcurrent PDs.
- 6) This protection scheme will continuously monitor the rated current magnitude at different PDs sites as demonstrated in (3) for fault detection. Upon fault detection, the fault location and type can be retrieved from the fault analysis data.
- 7) Based on the operating time of the PDs, the protection coordination status can be classified as being held or lost for a fault in a protection zone. If the coordination status is held, then permissible DG capacity can also be specified for these protection settings that do not require modifications of the protection settings.
- 8) To restore the protection coordination among PDs, the TDS of the DSDR can be revised in certain steps specified by the factor $\frac{I_{f,Rec}}{I_{f,Fuse}}$ to revise the fast and delayed operation of the reclosers. This will increase the coordination range to restore the fuse-saving scheme for the reliable operation of the protection network.
- 9) If the extreme TDS limits for relaying PDs have been reached without the restoration of protection coordination, the fuse size needs to be revised in these cases.
- 10) For the series fuses, (9) can be helpful to calculate the suitable characteristics of the new pairs of the series fuses to successfully establish the fuse–fuse coordination.
- 11) After the restoration of the recloser–fuse and fuse–fuse coordination, the permissible DG capacity can be specified for which this adaptive protection strategy is valid. If the protection coordination is successfully restored for all the possible scenarios, then no further action is required.

To meet the load growth in future grid expansions, the adaptive protection strategy should be reviewed after embedding a new DG unit. The proposed protection strategy by using dual-setting PDs will require a fewer number of PDs as DSDR can operate independently in both directions.

IV. RESULTS AND DISCUSSIONS

The validation of the proposed adaptive protection strategy has been performed by its implementation in DigSILENT PowerFactory and MATLAB on two different IEEE standardized test feeders to demonstrate its effectiveness and network reliability improvements. To analyze the different outcomes, the further discussion about the implementation of the proposed protection scheme on standardized DNs is described as follows.

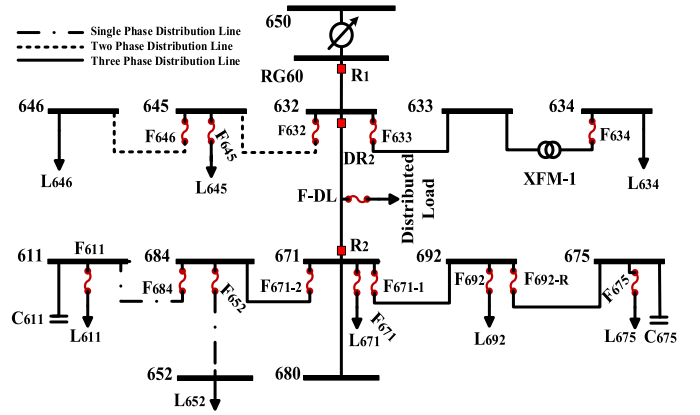


Fig. 6. IEEE 13 node test feeder along with PD placements.

TABLE I
SHORT-CIRCUIT PARAMETERS OF THE SBDG

Parameter	Value	Parameter	Value
X_l	0.05 pu	X_q	1.372 pu
R_a	0.0014 pu	$X'q$	0.8 pu
X_d	1.4 pu	$X''q$	0.118 pu
X'_d	0.231 pu	$T'q0$	1.25 s
X''_d	0.118 pu	$T''q0$	0.19 s
T'_{do}	5.5 s	$M=2H$	1.5 s
T''_{do}	0.05 s		

A. IEEE 13-Nodes Test Feeder

The analysis of the proposed protection strategy has been carried out by its implementation on IEEE 13 nodes test feeder due to its close resemblance with the practical DNs [31], [32]. The detailed information of this distribution test feeder is presented in [33]. The single line diagram of this test system is shown in Fig. 6 with optimal placement of the PDs as assessed by the authors in an earlier article presented in [8] with the requisite modifications. To analyze the impacts of the DG deployments on the conventional recloser–fuse coordination, a synchronous-based DG (SBDG) with the ratings of 4.05 MVA/0.69 kV via a step-up transformer (0.69/4.16 kV) with a leakage reactance of 0.15 p.u. is connected to the node 692. The impedance parameters of the SBDG are given in Table I for the short-circuit analysis. The nominal current ($I_{nom,i}$) for each i th node can be computed through conducting the unbalanced load flow for the test system as presented in Table II. The OLF was considered to be 1.25 for determining $I_{p,k}$ of the k th PDs at different nodes as described in Section II [34], [35].

1) *Protection Settings*: To define the short-circuit capability of all the PDs, the $I_{f,min}$ and $I_{f,max}$ values of each branch flow should be determined for each site as discussed in (5). The value, $I_{f,max}$, can be determined by simulating a three-phase short-circuit downstream of the associated PD location with negligible or no-fault impedance as shown in Table II. Similarly, $I_{f,min}$ can be computed by locating a single-line-to-ground

TABLE II
MAXIMUM/MINIMUM/RATED BRANCH CURRENT FOR IEEE 13-NODE
RADIAL FEEDER

Node		I_{nom}	$I_{f,min}$	$I_{f,max}$
From	To	(A)	(kA)	(kA)
RG60	632	587.1	1.24	5.41
632	633	81.1	0.72	6.73
632	645	143.3	0.93	4.25
632	671	478.2	1.08	4.52
XFM1-HV side		81.1	0.73	6.48
XFM1-LV side		704.7	0.59	18.76
645	646	64.9	0.84	3.64
671	692	229.3	0.78	3.69
671	684	70.9	0.69	3.47
671	680	0.0	0.59	6.25
692	675	205.9	0.71	7.83
684	611	70.8	0.63	1.93
684	652	63.1	0.66	2.67

(SLG) fault at the farthest node as listed in Table II. Based on the level of the operating voltage of the system under consideration, the requisite fault impedance value was considered to be 3Ω to calculate $I_{f,min}$ based on the information of the protection settings provided in [36]. These periphery limits of the fault current ensure the steadfast operation of the designated PDs for all fault current values in this range.

Based on this information, the protection settings of this test feeder can be carried out by utilizing the derived mathematical expressions in Section II-A for the entire fault current variation range. The starting feeder recloser R_1 was designed to exhibit the extremely inverse tripping characteristics from the series IAC77B801A of General Electric (GE) with the $TDS_{1,F}^{fw}$ of 0.5 and $TDS_{1,D}^{fw}$ of 10 to accommodate the series fuse coordination.

Furthermore, the fuse coefficients a_i and b_i can be calculated by utilizing the recloser operating time for the fast and delayed operation. The fuse coefficient a_i is chosen as -1.8 for distribution fuses as indicated in [12], [25], which represents the slope of the TCC curve of the fuses. This coefficient should be similar for all the fuses in a DN to establish appropriate series fuses coordination.

Moreover, the coefficient b_i has been computed using the derived formulation from (6) to (9) based on the fast and delayed operation time of the recloser as shown in Table III. For the downstream located fuses, the $t_{2,l,F}^{fw}$ and $t_{2,l,D}^{fw}$ of mid-line recloser R_2 will be utilized to compute the fuse coefficient b_i . Similarly, the mid-line recloser R_2 from the series of CDG34 of GE Alstom was opted to follow the extremely inverse TCC characteristics for the reclosers' mutual coordination with a minimum time delay of 10 cycles. The mutual coordination among reclosers is necessary to prevent the temporary interruption to the loads such as L645–L646 against downstream temporary faults.

The fuse coefficient b_i for all the downstream fuses was determined based on the operating times of R_2 by taking into account the series fuse arrangements in the faulted path as

TABLE III
FUSE COEFFICIENTS (a_i, b_i) FOR RECLOSER–FUSE COORDINATION

Fuse (i)	b_i	Fuse (i)	b_i
F632	6.38	F671	6.08
F633	6.83	F671-1	5.85
F634	8.32	F671-2	5.87
F645	6.65	F652	6.17
F646	6.67	F684	6.15
F-DL	6.54	F675	6.19
F611	6.19	F692-R	6.14
F692	6.13	----	----

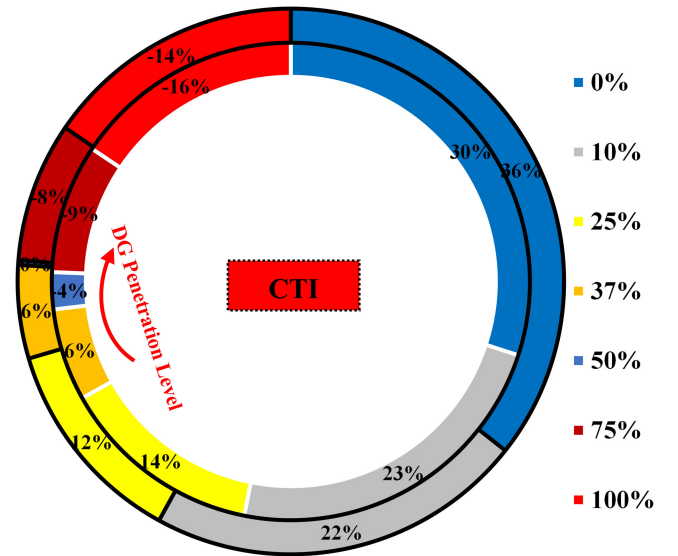


Fig. 7. Percentage variation of CTI with increasing DG penetration level at nodes 633 (internal) and 671 (external) for recloser–fuse coordination.

demonstrated in (9). For instance, for a fault current of 4219 A downstream of the fuse F645, the values of $t_{1,l,F}^{fw}$ and $t_{1,l,D}^{fw}$ are computed to be 0.097 and 1.932 s, respectively. Now, (9) can be used to calculate the fuse coefficient b_i to be 6.65 for the proper recloser–fuse coordination purposes. Similarly, all fuse coefficients can be computed for the primary recloser–fuse coordination purposes.

Furthermore, the fuses installed on the low-voltage side of the transformer are not necessarily coordinated with the reclosing PDs because these fuses are operating at different voltage levels and most of the transformer faults are permanent in nature. Therefore, the unsuccessful reclosing attempts against a permanent fault may originate serious reliability and stability issues.

2) *Results Analysis*: The preset recloser–fuse coordination may degrade with the increase of DG penetration level or by the proliferation of new DG units as has been exemplified in Fig. 7 with the percentage degradation of the CTI for recloser–fuse coordination for two different cases. The fault current experienced by the PDs varies significantly as the DG penetration level increases, which can result in nuisance tripping or protection

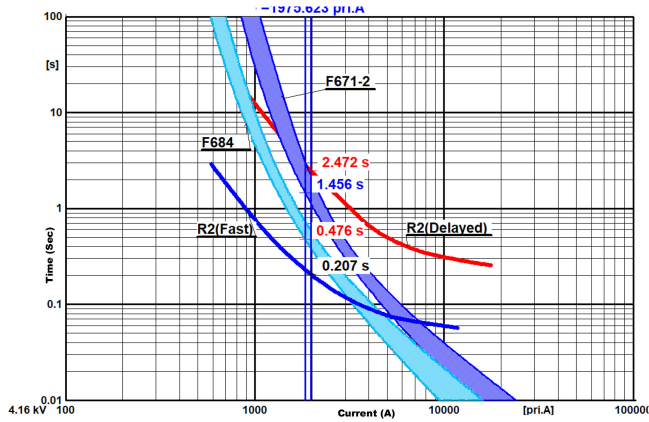


Fig. 8. Operation sequence of the PDs for a single-phase to ground (SLG) fault at node 611 to check the recloser–fuse coordination.

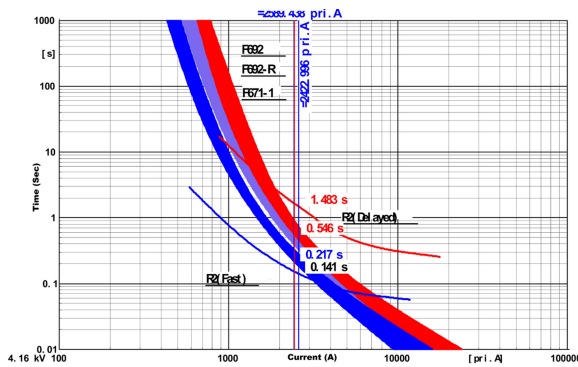


Fig. 9. Operation sequence of the PDs for double phase to ground (LLG) fault at the mid of 692 and 675 for a fault impedance of 1.0 Ω .

malfunctioning. Moreover, this impact was observed to be quite similar for different chosen DG locations.

The recloser–fuse coordination was not efficiently valid for an approximately 40% or higher DG penetration level for both cases. This discussion suggests that the protection settings among PDs need to be reviewed after the addition of new DG units to the power networks. This would be beneficial in ensuring the safe operation of the utility network to accommodate future load growths. Furthermore, it will be beneficial to improve the reliability metrics by the successful implementation of the fuse-saving scheme.

Now, the operation sequence of the recloser and fuses was validated for the implementation of the fuse saving scheme as demonstrated in Fig. 8 for an SLG fault at node 611, which is the most common power network fault [37]. The fast mode of R_2 was set to operate at $t = 0.207$ s for self-clearing the temporary faults. The fuse F684 was set to reach its melting point at $t = 0.476$ s if the fault was not cleared by fast operation of R_2 , backed by the series fuse F671-2 in the faulted path set to clear the fault at $t = 1.456$ s. However, if both fuses fail to clear the fault, the recloser R_2 will tend to be locked out permanently at $t = 2.472$ s.

Furthermore, Fig. 9 is illustrating the similar recloser–fuse coordination for the fuses installed on the laterals 671–675 by locating a double phase to ground fault (LLG) at the middle

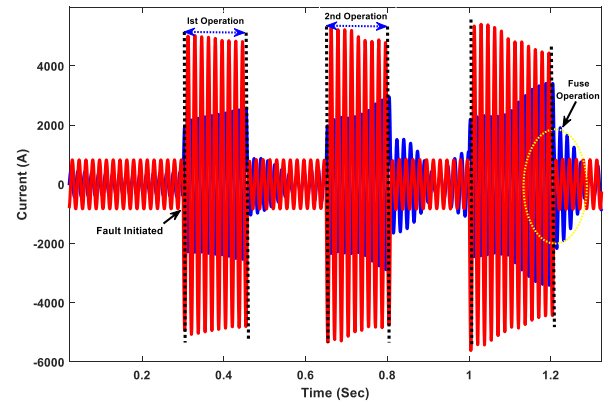


Fig. 10. Time-domain fault current magnitude at node 632 for line–line (a–c) fault for a fault impedance of 0.2 Ω .

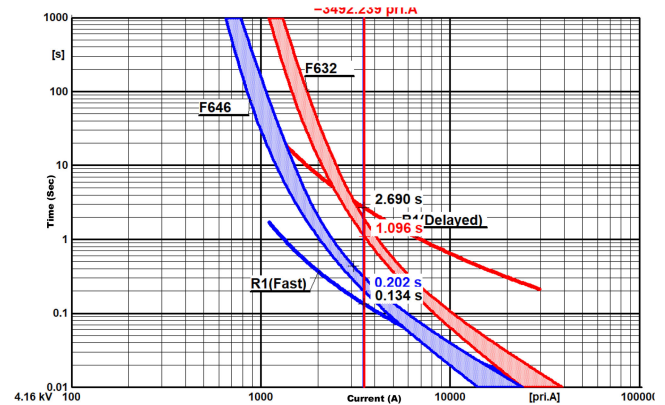


Fig. 11. Operation sequence of the PDs for line–line (LL) fault at node 646 with a fault impedance of 1 Ω .

of the node 692 and 675. The fuse–fuse coordination can also be observed among the series fuses, i.e., F671-1 and F692-R. The fast and delayed operation of recloser R_2 demonstrates the effective implementation of the fuse-saving protection strategy.

Based on the electric utility reliability standards, the network engineer will set the number of the recloser fast trips before being permanently locked out. The fuse melting should begin after the execution of these reclosing attempts as demonstrated in Fig. 10 for a line-to-line (LL) fault at node 684. The time-domain plot of the fault current at node 632 illustrates the execution of the two reclosing attempts under the fast mode before the fuse operation at $t = 1.21$ s thus removing the faulted section. The fault clearance by the associated fuse will improve the selectivity of the protection network.

Similarly, feeder recloser R_1 was set to exhibit mutual coordination for the fuses F632 and F646 in its protection zones and downstream-located PDs as shown in Fig. 11 for an impedance LL fault at node 646. The sequence of operation of different PDs clearly shows the proper coordination for the fuse-saving scheme.

Furthermore, it should be ensured that both the reclosers R_1 and R_2 have the mutual coordination for the downstream located PDs for the improved selectivity of the protection network. However, the original protections settings do not guarantee the

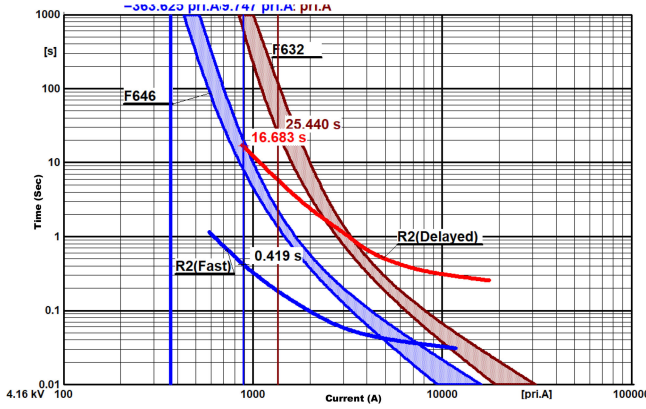


Fig. 12. Operation sequence of the PDs for line-to-line (LL) fault at node 645 with a fault impedance of 1.5Ω .

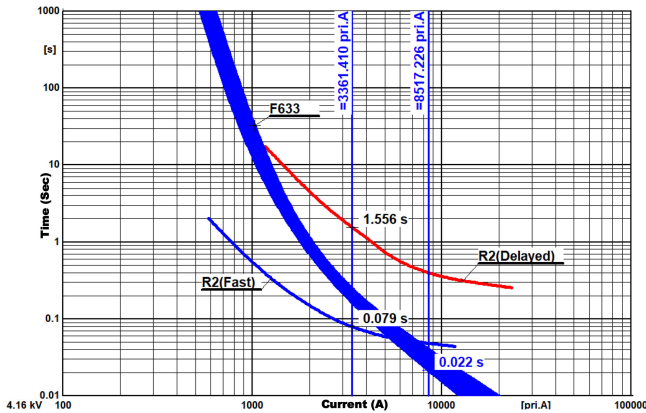


Fig. 13. Operation sequence of the PDs for a symmetrical fault at 10% of the total line between nodes 632 and 633.

requisite protection coordination among the same fuses F632 and F646 with the recloser R_2 in its reverse operation in the presence of DG as shown in Fig. 12 for an LL fault at node 645.

A similar coordination issue can be observed for a symmetrical fault at 10% of the total line section between nodes 632 and 633 to investigate the fuse-saving scheme as shown in Fig. 13. It can be seen that the fuse will operate before the fast operation of R_2 without giving a chance for the self-clearance of the momentary faults. This protection maloperation was due to the dissimilar fault level experienced by both recloser and fuse. The identical protection settings for R_2 may not be valid for all the fault locations in the DN.

Therefore, it can be concluded that the nuisance operation of these PDs will originate from the undesirable islands in the power network thus deteriorating the network reliability indices. Consequently, the conventional protection settings of the PDs may not be valid to establish proper recloser–fuse operation for all the fault events. Therefore, the DSDR can be utilized to provide the autonomous protection settings in both directions irrespective of the fault level experienced. The classification of the existence of the protection coordination has been demonstrated by Fig. 14 for all the nodes against different types of fault events. The protection coordination is classified as lost if the fuse-saving

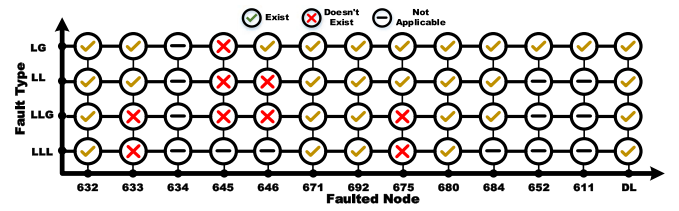


Fig. 14. Classification of the protection coordination for all the nodes against different fault types without the DSDR.

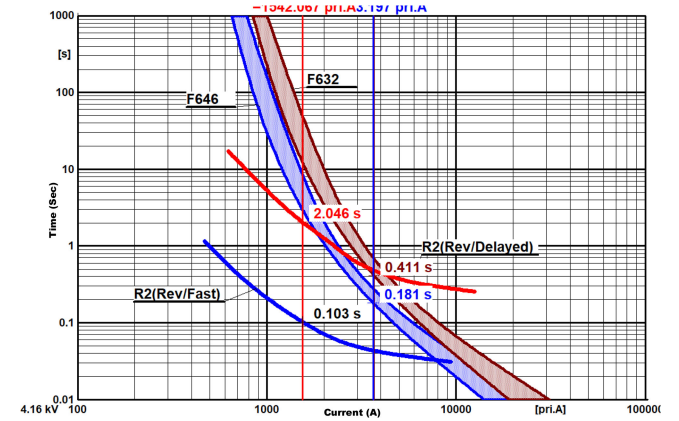


Fig. 15. Fault level experienced by each PD and their operation sequence for solid line–line (LL) fault at node 646.

scheme cannot be implemented successfully during a particular fault event.

It can be observed that chances of coordination loss are higher when the fault location is at the upstream of the recloser R_2 thus signifying the necessity of DSDR for the protection system of modern DNs. This discussion will lead to the replacement of the mid-line recloser R_2 with dual-setting reclosing PD having independent protection settings in both directions. It will enable a flexible protection range for the recloser–fuse and fuse–fuse coordination. The characterization enhancement of the PDs can reduce the number of devices required thus simplifying the design of the protection scheme.

Based on these recommendations, the reclosers R_2 can be modified to exhibit the dual-setting characteristics for the forward and reverse operation. In the forward direction, similar characteristics can be utilized as it shows the valid coordination for the majority of the cases. Furthermore, the factor $\frac{I_{f,Rec}}{I_{f,Fuse}}$ can be utilized to precisely revise the TCC curves of the reclosing PD to restore the lost protection coordination. In the reverse direction, the basic features will also be based on the series CDG34 of GE Alstom with extremely inverse operating characteristics. The protection settings will be diverse for each direction of operation as demonstrated in Fig. 15 for a solid LL fault at node 646. The fault current experienced by the series fuse is 3.64 kA as the summation of individual fault current contribution from the grid and DG source. However, the fault current experienced by the reclosers R_2 is considerably less, i.e., 1.54 kA, because it will only experience fault current contribution from DG source. Thus, the DSDR can cope with the different protection settings in modern power grids for dissimilar fault levels.

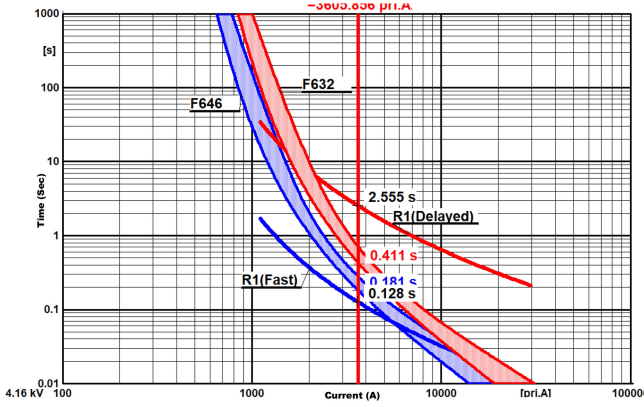


Fig. 16. Operation sequence of the PDs for solid line-to-line fault at node 646.

TABLE IV
VALIDATION OF RECLOSER-FUSE COORDINATION FOR DIFFERENT FAULTS

Faulted Node	R_1		R_2		Fuse ($t_{fuse(i)}^{MMT}$)
	Fast	Delayed	Fast	Delayed	
632	0.070	1.398	0.045	0.514	---
633	0.094	1.880	0.052	0.706	0.163
645	0.105	2.095	0.087	1.659	0.276
646	0.128	2.555	0.103	2.046	0.181
DL	0.087	1.733	0.076	1.402	0.248
671	0.105	2.091	0.068	1.113	---
692	0.115	2.307	0.072	1.258	0.175
675	0.121	2.414	0.074	1.332	0.185
684	0.125	2.507	0.075	1.394	0.205
680	0.137	2.747	0.079	1.555	---
611	0.326	6.523	0.146	4.385	0.417
652	0.676	13.525	0.143	9.370	0.233

Moreover, the operating sequence of these PDs shows the successful implementation of the fuse-saving scheme for self-clearing of the momentary faults. The fault current contribution from each source will be interrupted by the recloser R_1 and R_2^{rv} fast operations at $t = 0.128$ s and $t = 0.103$ s, respectively, as shown in Figs. 15 and 16.

However, if the fault is not temporary in nature, the fuse F646 will start its operation at $t = 0.181$ s followed by the operation of the main lateral fuse F632 at $t = 0.411$ s. The successful fuse operation will eventually clear the faulted section thus preventing the reclosers R_1 and R_2^{rv} lockout operations set to operate at $t = 2.555$ s and $t = 2.046$ s, respectively. If the conventional protection settings for R_2 has been used, the loss of recloser-fuse coordination could have resulted in undesirable islands in the DN. Furthermore, the operation time constraints of different PDs are presented in Table IV for different fault events for an overview of the proposed protection strategy. It can be observed that both reclosers have mutual coordination along with the series fuses in the faulted path to minimize the interruptions.

Based on the proposed protection strategy, the validation of the protection coordination can also be presented as shown in Fig. 17 for different types of fault events to effectively implement

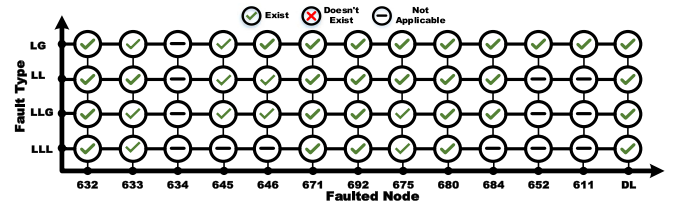


Fig. 17. Classification of the protection coordination for all the nodes against different fault types by utilizing the proposed protection strategy with DSDR.

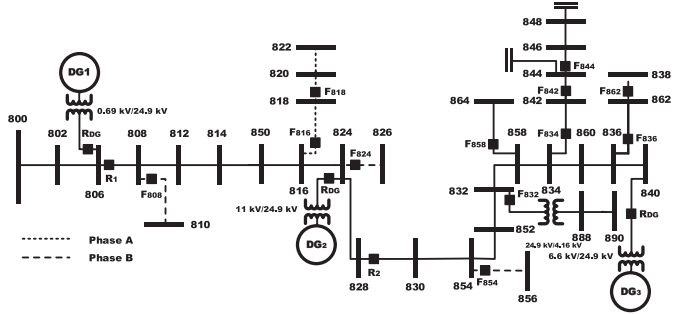


Fig. 18. Single diagram of modified IEEE 34-node test feeder with DGs.

the fuse-saving scheme. The deployment of DSDR will enable the network operators to completely de-energize any faulted section for the self-clearance of the temporary faults. The steadfast elimination of the momentary faults will subsequently prevent sustained outages to improve the network reliability.

Furthermore, the enhanced flexible coordination range associated with DSDR will be considerably beneficial for the lengthy DNs with hundreds of PDs to be mutually coordinated. The perfect coordination among the reclosing PDs will prevent momentary interruptions to a larger number of consumers. Moreover, it will increase the range of the protection zones associated with DSDR in the network. Additionally, it will prevent undesirable islanding in the power network thus improving the reliability indices of the network.

B. IEEE 34-Nodes Test System

To demonstrate the effectiveness of the proposed protection strategy, the IEEE 34 node test system [33] has also been opted to validate the proposed scheme as shown in Fig. 18 along with the PDs allocation. Furthermore, three DGs were located at different nodes, i.e., DG₁ (750 KVA/0.69 kV) at node 806, DG₂ (500 KVA/11 kV) at node 824, and DG₃ (600 KVA/6.6 kV) at node 840. The start-line recloser R_1 was set to follow the extremely inverse characteristics for the IAC77M5A series of GE for its fast and delayed operations. Similarly, the mid-line recloser R_2 was selected from the IAC78A801A series of GE to exhibit the extremely inverse characteristics for both fast and delayed operations for coordination purposes. To implement the conventional fuse-saving scheme, all the PDs were mutually coordinated using the procedural steps discussed in the previous sections. Furthermore, both reclosing PDs were mutually coordinated along with the fuses in their protection zones.

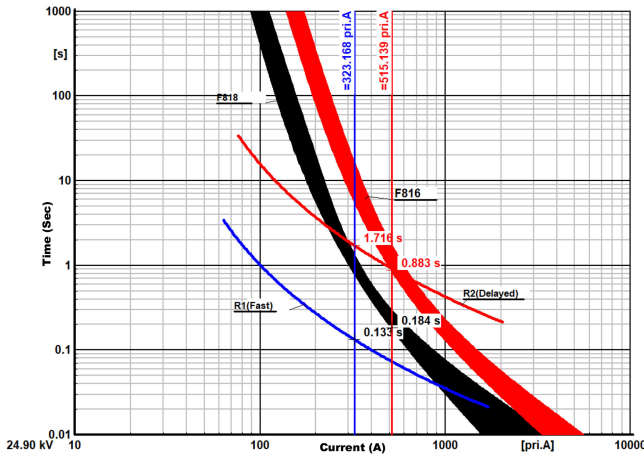


Fig. 19. Fault level experienced by each PD and their operation sequence for solid line-to-ground (LG) fault at node 820.

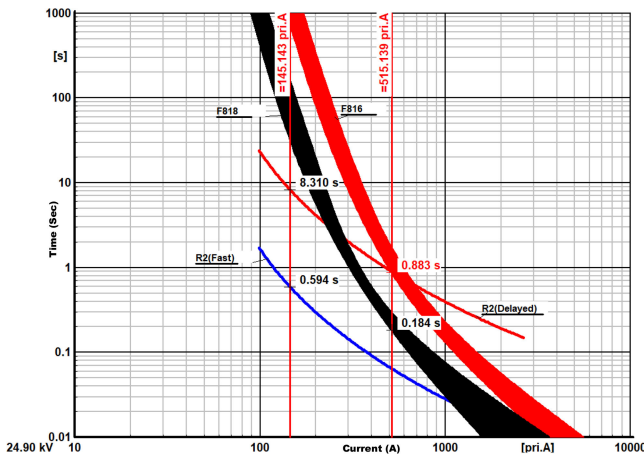


Fig. 20. Nuisance tripping of the conventional PDs for solid line-to-ground (LG) fault at node 820.

The recloser R_1 was coordinated with the downstream lateral fuses in its protection zones as demonstrated in Fig. 19 for an SLG fault at node 820. Although recloser R_1 and both lateral fuses (F_{816} and F_{818}) experience different fault currents due to the DGs presence, the recloser fast and delayed operation was also coordinated with the series lateral fuses.

However, the conventional recloser R_2 does not exhibit the proper coordination in the reverse direction for the same SLG fault at node 820 because of the significant diverse fault current experienced in the forward and reverse directions. As shown in Fig. 20, it can be observed that fast operation of reclose R_2 will be tardy to initiate at $t = 0.594$ s for a reduced fault current of 145.143 A from DG_3 in comparison to the associated fuse F_{818} , which is likely to operate at $t = 0.184$ s for a fault current of 515.139 A from all the sources. This will result in the loss of the fuse-saving scheme as lateral fuses may operate prior to the recloser fast operation without any probability of the self-clearance of temporary faults thus reducing network reliability.

Thus, the conventional protection settings for recloser R_2 may not be appropriate for all fault scenarios. Therefore, recloser R_2 can be modified to exhibit the DSDR protection characteristics

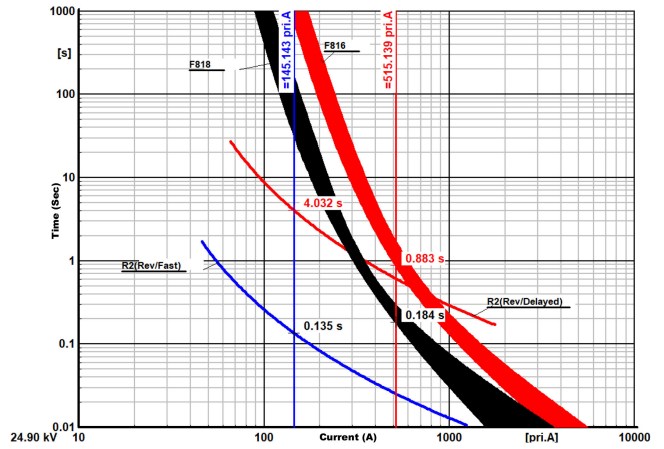


Fig. 21. Operation sequence of the PDs for solid line-to-ground (LG) fault at node 820.

for forward and reverse characteristics to successfully implement the fuse saving scheme for all the fault events. In the reverse direction, the extremely inverse protection settings from the similar IAC78A801A series of GE can be chosen for the recloser R_2 to establish the perfect recloser–fuse coordination. The deployment of DSDR will enable the independent pickup and TDS protection settings in both directions irrespective of the experienced fault levels.

For the same SLG fault at node 820, the modified protection settings of recloser R_2^{rv} will enable to interrupt the fault current contribution from DG_3 by its fast operation at $t = 0.135$ s in the reverse direction as shown in Fig. 21. The interruptions of the fault current contributions by associated PDs will originate the probability of self-clearance of the momentary faults. If the fault was persistent in nature, the associated fuse F_{818} will clear the faulted section to minimize the power disruptions. Furthermore, the fuse F_{816} and delayed operations of both reclosing PDs will constitute secondary and tertiary layers of the protection network.

The deployment of dual-setting directional PDs will prevent the undesirable islands in the power networks by averting the nuisance trippings of protection system due to multiple DGs incorporation. Similarly, the effectiveness of the proposed protection scheme can also be assessed by observing the protection coordination constraints for different faults events as shown in Table V. It can be seen for different fault events that the fuse-saving scheme is implemented successfully for different fault locations. Furthermore, both the reclosers R_1 and R_2 have mutual coordination in addition to the associated fuse to minimize the network outages.

In addition, the mutual coordination among the reclosing PDs will minimize the number of customers experiencing the momentary supply interruptions due to the operation of the associated recloser only. Similarly, the autonomous protection settings of the DSDR will enable the successful implementation of the fuse-saving scheme for all the fault locations in the large DNs. These factors will play a vital role in the improvements of the reliability indices of the lengthy DN along with an enhanced flexible coordination range.

TABLE V
VALIDATION OF RECLOSER–FUSE COORDINATION FOR DIFFERENT FAULTS
EVENTS IN THE DISTRIBUTION NETWORK

Faulted Node	R_1		R_2		Fuse ($t_{fuse(i)}^{MMT}$)
	Fast	Delayed	Fast	Delayed	
810	0.031	0.368	0.171	5.303	0.267
814	0.088	1.105	0.077	2.120	---
818	0.089	1.128	0.078	2.165	0.330
820	0.133	1.716	0.135	4.032	0.184
824	0.099	1.256	0.071	1.929	---
832	0.190	2.515	0.139	1.943	---
842	0.211	2.809	0.155	2.170	1.015
844	0.214	2.862	0.158	2.210	0.459
846	0.224	3.007	0.166	2.321	0.248
836	0.219	2.926	0.161	2.260	---
838	0.228	3.053	0.168	2.347	0.263
852	0.190	2.515	0.139	1.943	---
856	0.156	2.032	0.112	1.563	0.519
860	0.214	2.853	0.157	2.204	---
862	0.220	2.937	0.162	2.268	0.658
864	0.201	2.676	0.148	2.068	0.590

V. CONCLUSION

The large-scale integration of the DERs may cause considerable degradation to the network reliability indices. The recloser–fuse coordination is among the most vulnerable protection issues of the DNs equipped with multiple DGs. This article focuses on the design of an adaptive protection strategy based on dual-setting reclosing PDs to provide a substantial flexible coordination range for recloser–fuse and fuse–fuse coordination purposes with the help of extensive mathematical formulations. The utilization of DSDR will provide a flexible coordination range to successfully implement fuse-saving scheme to prevent permanent outages due to momentary faults. Furthermore, the proposed protection scheme will facilitate the network operator to utilize fewer numbers of PDs to protect the entire power grid. The proposed protection strategy has been validated using the IEEE standardized test feeders for different types of fault events. A comparative analysis has also been presented to demonstrate the significance of the proposed protection strategy in comparison with those of the conventional protection practices.

REFERENCES

- [1] H. Nasiri, S. Bahrnamirad, M. Shahidehpour, A. E. Paaso, N. M. Abdullah, and D. Kushner, "Networked-constrained DER valuation in distribution networks," *IEEE Trans. Smart Grid*, vol. 11, no. 6, pp. 4809–4817, Nov. 2020.
- [2] C. Galvez and A. Abur, "Fault location in active distribution networks containing distributed energy resources (DERs)," *IEEE Trans. Power Del.*, vol. 36, no. 5, pp. 3128–3139, Oct. 2021.
- [3] R. Walling, R. Saint, R. C. Dugan, J. Burke, and L. A. Kojovic, "Summary of distributed resources impact on power delivery systems," *IEEE Trans. Power Del.*, vol. 23, no. 3, pp. 1636–1644, Jul. 2008.
- [4] A. Yazdanejadi, S. Golshannavaz, D. Nazarpour, S. Teimourzadeh, and F. Aminifar, "Dual-setting directional overcurrent relays for protecting automated distribution networks," *IEEE Trans. Ind. Inform.*, vol. 15, no. 2, pp. 730–740, Feb. 2019.
- [5] S. A. Saleh *et al.*, "Solid-state transformers for distribution systems—part II: Deployment challenges," *IEEE Trans. Ind. Appl.*, vol. 55, no. 6, pp. 5708–5716, Nov./Dec. 2019.
- [6] A. Girgis and S. Brahma, "Effect of distributed generation on protective device coordination in distribution system," in *Proc. Powering Beyond Large Eng. Syst. Conf. Power Eng.*, 2001, pp. 115–119.
- [7] S. M. Brahma and A. A. Girgis, "Development of adaptive protection scheme for distribution systems with high penetration of distributed generation," *IEEE Trans. Power Del.*, vol. 19, no. 1, pp. 56–63, Jan. 2004.
- [8] M. Yousaf, K. M. Muttaqi, and D. Sutanto, "Overcurrent protection scheme for the IEEE 13-node benchmark test feeder with improved selectivity," in *Proc. IEEE Power Energy Soc. General Meeting*, 2020, pp. 1–5.
- [9] M.-H. Kim, S. Lim, J. Moon, and J.-C. Kim, "Method of recloser-fuse coordination in a power distribution system with superconducting fault current limiter," *IEEE Trans. Appl. Supercond.*, vol. 20, no. 3, pp. 1164–1167, Jul. 2010.
- [10] S. Santoso and T. A. Short, "Identification of fuse and recloser operations in a radial distribution system," *IEEE Trans. Power Del.*, vol. 22, no. 4, pp. 2370–2377, Oct. 2007.
- [11] S. Brahma and A. Girgis, "Microprocessor-based reclosing to coordinate fuse and recloser in a system with high penetration of distributed generation," in *Proc. IEEE Power Eng. Soc. Winter Meeting Conf.*, 2002, pp. 453–458.
- [12] S. Chaitusaney and A. Yokoyama, "Prevention of reliability degradation from recloser–fuse miscoordination due to distributed generation," *IEEE Trans. Power Del.*, vol. 23, no. 4, pp. 2545–2554, Oct. 2008.
- [13] B. Hussain, S. Sharkh, S. Hussain, and M. Abusara, "An adaptive relaying scheme for fuse saving in distribution networks with distributed generation," *IEEE Trans. Power Del.*, vol. 28, no. 2, pp. 669–677, Apr. 2013.
- [14] H. Zhan *et al.*, "Relay protection coordination integrated optimal placement and sizing of distributed generation sources in distribution networks," *IEEE Trans. Smart Grid*, vol. 7, no. 1, pp. 55–65, Jan. 2016.
- [15] M. Kim, S. Lim, and J. Kim, "Improvement of recloser-fuse operations and coordination in a power distribution system with SFCL," *IEEE Trans. Appl. Supercond.*, vol. 21, no. 3, pp. 2209–2212, Jun. 2011.
- [16] A. Pregelj, M. Begovic, and A. Rohatgi, "Recloser allocation for improved reliability of DG-enhanced distribution networks," *IEEE Trans. Power Syst.*, vol. 21, no. 3, pp. 1442–1449, Aug. 2006.
- [17] A. Conde and E. Vázquez, "Operation logic proposed for time overcurrent relays," in *Proc. IEEE Power Eng. Soc. General Meeting*, 2006, pp. 1–10.
- [18] S. Venkata, M. J. Reno, W. Bower, S. Manson, J. Reilly, and G. W. Sey Jr, "Microgrid protection: Advancing the state of the art," Sandia National Laboratories, Albuquerque, New Mexico, Rep. no. SAND2019-3167, 2019.
- [19] H. H. Zeineldin, H. M. Sharaf, D. K. Ibrahim, and E. E.-D. Abou El-Zahab, "Optimal protection coordination for meshed distribution systems with DG using dual setting directional over-current relays," *IEEE Trans. Smart Grid*, vol. 6, no. 1, pp. 115–123, Jan. 2015.
- [20] H. M. Sharaf, H. Zeineldin, and E. El-Saadany, "Protection coordination for microgrids with grid-connected and islanded capabilities using communication assisted dual setting directional overcurrent relays," *IEEE Trans. Smart Grid*, vol. 9, no. 1, pp. 143–151, Jan. 2018.
- [21] A. Yazdanejadi, D. Nazarpour, and S. Golshannavaz, "Dual-setting directional over-current relays: An optimal coordination in multiple source meshed distribution networks," *Int. J. Elect. Power Energy Syst.*, vol. 86, pp. 163–176, 2017.
- [22] M. Yousaf, A. Jalilian, K. M. Muttaqi, and D. Sutanto, "An adaptive protection scheme for dual-setting directional recloser and fuse coordination for distribution grids with distributed generation," in *Proc. IEEE Ind. Appl. Soc. Annu. Meeting*, 2021, pp. 1–7.
- [23] G. Benmouyal *et al.*, "IEEE standard inverse-time characteristic equations for overcurrent relays," *IEEE Trans. Power Del.*, vol. 14, no. 3, pp. 868–872, Jul. 1999.
- [24] M. Yousaf and T. Mahmood, "Protection coordination for a distribution system in the presence of distributed generation," *Turkish J. Elect. Eng. Comput. Sci.*, vol. 25, no. 1, pp. 408–421, 2017.
- [25] A. Naïem, Y. Hegazy, A. Abdelaziz, and M. Elsharkawy, "A classification technique for recloser-fuse coordination in distribution systems with distributed generation," *IEEE Trans. Power Del.*, vol. 27, no. 1, pp. 176–185, Jan. 2012.
- [26] M. N. Alam, B. Das, and V. Pant, "Optimum recloser–fuse coordination for radial distribution systems in the presence of multiple distributed generations," *IET Gener. Transmiss. Distrib.*, vol. 12, no. 11, pp. 2585–2594, 2018.

- [27] M. Yousaf, A. Jalilian, K. M. Muttaqi, and D. Sutanto, "An adaptive grounding scheme for synchronous-based DG to prevent the generator damage and protection malfunctioning during ground faults under different operating modes," *IEEE Trans. Ind. Appl.*, vol. 57, no. 3, pp. 2307–2316, May/Jun. 2021.
- [28] W. J. Ruschel and A. W. Ashley, "Coordination of relays, reclosers, and sectionalizing fuses for overhead lines in the oil patch," *IEEE Trans. Ind. Appl.*, vol. 25, no. 6, pp. 1041–1048, Nov./Dec. 1989.
- [29] R. P. Broadwater, J. C. Thompson, R. E. Lee, and H. Maghdan-D, "Computer-aided protection system design with reconfiguration," *IEEE Trans. Power Del.*, vol. 6, no. 1, pp. 260–266, Jan. 1991.
- [30] S. A. Saleh, C. Richard, X. F. S. Onge, J. Meng, and E. Castillo-Guerra, "Comparing the performance of protection coordination and digital modular protection for grid-connected battery storage systems," *IEEE Trans. Ind. Appl.*, vol. 55, no. 3, pp. 2440–2454, May/Jun. 2019.
- [31] M. Bahrampianah, D. Torregrossa, R. Cherkaoui, and M. Paolone, "A decentralized adaptive model-based real-time control for active distribution networks using battery energy storage systems," *IEEE Trans. Smart Grid*, vol. 9, no. 4, pp. 3406–3418, Jul. 2018.
- [32] S. Paudyal, C. A. Canizares, and K. Bhattacharya, "Optimal operation of distribution feeders in smart grids," *IEEE Trans. Ind. Electron.*, vol. 58, no. 10, pp. 4495–4503, Oct. 2011.
- [33] W. H. Kersting, "Radial distribution test feeders," *IEEE Trans. Power Syst.*, vol. 6, no. 3, pp. 975–985, Aug. 1991.
- [34] D. Birla, R. P. Maheshwari, and H. O. Gupta, "A new nonlinear directional overcurrent relay coordination technique, and banes and boons of near-end faults based approach," *IEEE Trans. Power Del.*, vol. 21, no. 3, pp. 1176–1182, Jul. 2006.
- [35] W. El-Khattam and T. Sidhu, "Restoration of directional overcurrent relay coordination in distributed generation systems utilizing fault current limiter," *IEEE Trans. Power Del.*, vol. 23, no. 2, pp. 576–585, Apr. 2008.
- [36] V. De Andrade and E. Sorrentino, "Typical expected values of the fault resistance in power systems," in *Proc. Transmiss. Distrib. Conf. Expo.: Latin Amer.*, 2010, pp. 602–609.
- [37] S. Saleh, A. Aljankawey, R. Meng, J. Meng, L. Chang, and C. Diduch, "Impacts of grounding configurations on responses of ground protective devices for DFIG-based WECSs—Part II: High-impedance ground faults," *IEEE Trans. Ind. Appl.*, vol. 52, no. 2, pp. 1204–1214, Mar./Apr. 2016.



Muhammad Yousaf (Student Member, IEEE) received the B.Sc. and M.Sc. degrees in electrical engineering from the University of Engineering and Technology, Taxila, Pakistan, in 2012 and 2015, respectively. He is currently working toward the Ph.D. degree in electrical engineering at the School of Electrical, Computer and Telecommunication Engineering, University of Wollongong, Wollongong, NSW, Australia.

He was a Lecturer of Electrical Engineering with the University of Gujrat, Punjab, Pakistan from 2015 to 2018. He is currently a Research Student with the Australian Power Quality and Reliability Centre, University of Wollongong, Wollongong, NSW, Australia. He is also a Reviewer for IEEE TRANSACTIONS ON INDUSTRY APPLICATIONS, *IET Generation, Transmission & Distribution*, and IEEE ACCESS, etc. His research interests include power system protection, distributed generation, microgrids, grid integration of renewable energy resources, and power system grounding.



Amin Jalilian (Student Member, IEEE) received the M.Sc. degree in electrical power engineering from the University of Tabriz, Tabriz, Iran, in 2014. He is currently working toward the Ph.D. degree in electrical engineering at the University of Wollongong, Wollongong, NSW, Australia.

Before starting the Ph.D. studies, he was involved with Power Industry for more than six years, where he worked as a Consultant and Supervision Engineer in the field of electric power transmission and distribution. His research interests include the renewable energy integration, power system protection, dynamics, and stability.



Kashem M. Muttaqi (Senior Member, IEEE) received the B.Sc. degree in electrical and electronic engineering from the Bangladesh University of Engineering and Technology, Dhaka, Bangladesh, in 1993, the M.Eng.Sc. degree in electrical engineering from the University of Malaya, Kuala Lumpur, Malaysia, in 1996, and the Ph.D. degree in electrical engineering from Multimedia University, Selangor, Malaysia, in 2001.

He is currently a Professor with the School of Electrical, Computer, and Telecommunications Engineering, University of Wollongong, Wollongong, NSW, Australia. He was associated with the University of Tasmania, Hobart, TAS, Australia as a Research Fellow/Lecturer/Senior Lecturer from 2002 to 2007, and with the Queensland University of Technology, Brisbane, QLD, Australia, as a Research Fellow from 2000 to 2002. Previously, he also worked with Multimedia University as a Lecturer for three years. He has more than 21 years of academic experience and authored or coauthored more than 400 research papers in international journals and conference proceedings. His research interests include distributed generation, solid-state transformer, renewable energy resources, electrical vehicles, smart grid, power system planning, and distribution networks.



Danny Sutanto (Senior Member, IEEE) received the B.Eng. (Hons.) and Ph.D. degrees in electrical engineering from the University of Western Australia, Perth, WA, Australia, in 1978 and 1981, respectively.

He joined the University of New South Wales, Sydney, NSW, Australia, in 1982, and Hong Kong Polytechnic University, Hong Kong, in 1996, as a Professor of Electrical Engineering. Since 2006, he has been a Professor of power engineering with the University of Wollongong, Wollongong, NSW, Australia. He has authored or coauthored more than

400 research papers in international journals and conference proceedings. His research interests include power system planning, power system emergency, analysis and harmonics, flexible alternating current transmission system, and battery energy storage systems.

Prof. Sutanto was the IEEE Industry Applications Society Area Chair for Region 10 (Asia Pacific) from 2014 to 2017.